

Tianchu Zeng

PhD Candidate, Electrical & Computer Engineering · National University of Singapore

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PROFESSIONAL SUMMARY

PhD candidate in Electrical & Computer Engineering at the National University of Singapore working in computational neuroscience and AI for science, with a focus on biomedical research. Research develops machine-learning and statistical methods to accelerate biophysical brain-model fitting, improve the statistical validity of machine-learning evaluation, and study brain foundation models. Experienced in leading interdisciplinary projects from method design and large-scale experiments to reproducible workflows and open-source software.

EDUCATION

National University of Singapore — Singapore

Aug 2022 - Present (expected 2027)

Ph.D. in Electrical & Computer Engineering

- Advisor: Assoc Prof. BT Thomas Yeo
- Co-advisor: Prof. Thomas E. Nichols, University of Oxford
- Thesis: Machine Learning in Brain Imaging
- GPA: 4.88 / 5.0
- Graduate coursework: Pattern Recognition, Visual Computing, Deep Learning, and Information Theory (all A/A-)

Tsinghua University — Beijing, China

Aug 2018 - Jun 2022

B.Eng. in Department of Automation

- GPA: 3.74 / 4.00
- Thesis: Research on Trajectory Prediction Methods for Traffic Agents in Mixed Traffic Environments
- Advisor: Prof. Jianming Hu & Prof. Yi Zhang

RESEARCH EXPERIENCE

Machine Learning in Brain Imaging — National University of Singapore

Aug 2022 - Present

- Developed a deep-learning surrogate that accelerated fitting of biophysical brain models by 50–100× while producing comparable parameter estimates; first-author work accepted in principle at *Nature Methods*.
- Led the end-to-end modeling and validation workflow and coordinated its extension to 12,005 participants across 14 datasets to estimate cortical excitation-inhibition balance.
- Developed a statistically valid method for comparing machine-learning models under repeated cross-validation; led a review of 210 biomedical studies showing that 97% used tests that ignored dependence between folds, and released a scikit-learn-compatible implementation.
- Contributed to benchmarking brain foundation models against classical approaches for phenotypic prediction and helped shape the lab's foundation-model research direction.

Brain Parcellation with Resting-State fMRI — National University of Singapore

Aug 2022 – Jun 2023

- Developed methods for group- and individual-level brain parcellation using resting-state fMRI.
- Performed functional-connectivity analysis and cortical-surface visualization across volume and surface data using MATLAB and FreeSurfer/Freeview.

Trajectory Prediction with Graph Neural Networks — Tsinghua University

Jan 2020 - Jun 2022

- Built RNN-, GNN-, CVAE-, and Transformer-based models for multi-agent trajectory prediction and constructed a simulated Shougang Industrial Park dataset.
- Work resulted in two peer-reviewed conference papers and two Chinese patent applications.

DNA Encryption/Decryption System — Tsinghua University

Sep 2020 - Jun 2021

- Led an iGEM project developing a DNA encryption/decryption system with heuristic decoding for binary data encoded into base sequences.

SELECTED WORKS

* denotes equal contribution/co-first author; + denotes co-corresponding author; author name in bold.

Selected Publications

1. **Zeng, T.***, Tian, F.*, Zhang, S., Li, X., Tan, A. P., Larsen, B., ... & Yeo, B. T. (2026). Optimizing biophysical large-scale brain circuit models with deep neural networks. *Nature Methods*. Accepted in principle.
2. Qu, Y. L., Chen, J., Tam, A., Ooi, L. Q. R., Dhamala, E., Cocuzza, C. V., Zhang, S., **Zeng, T.**, ... & Holmes, A. J. (2025). Distinct brain network features predict internalizing and externalizing traits in children, adolescents and adults. *Nature Mental Health*, 1-12.
3. Zhang, S.*, Larsen, B.*, Sydnor, V. J.*, **Zeng, T.**, An, L., Yan, X., ... & Yeo, B. T. (2024). In vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *Proceedings of the National Academy of Sciences*, 121(23), e2318641121.

Selected Preprints & Manuscript in preparation

1. **Zeng, T.***, Li, H.*, Zhang, S.*, Tan, Y.Q., Tian, F., Orban, C., ..., Nichols, T.+, & Yeo, B. T.+ (2026). Widespread use of invalid statistical tests in biomedical machine learning. [bioRxiv](#) preprint.
 2. Ling, Z., Dong, Z., **Zeng, T.**, Lin, Y., Li, R., Che, W., Zhang, C., Holmes, A., An, L., Zhang, S., Zhou, J. H., & Yeo, B. T. (2026). *Do Brain Foundation Models Perform Better in Phenotypic Prediction than Classical Models?* Manuscript in preparation; preprint forthcoming.
- Also co-first authored two peer-reviewed transportation conference papers (CICTP 2022; ICTD 2022).

SELECTED SOFTWARE & OPEN-SOURCE PROJECTS

admissions-ops — Creator and Maintainer

- Designed and open-sourced a local-first AI-agent workspace that turns official program pages and applicant evidence into sourced evaluations, eligibility matrices, tailored CVs/resumes, and review-required statement drafts.
- Architected an "LLM authors, deterministic layer guards" workflow using reusable agent skills for Codex and Claude Code, structured file contracts, provenance tracking, and 13 CI validation targets.

Whackamon — Creator

- Designed, built, and deployed a bilingual browser game with seven progression levels, dynamic difficulty, achievements, persistent local state, audio, and mobile packaging using TypeScript, React, Phaser, Vite, Zustand, and Capacitor.
- Published on whackamon.com and itch.io

INDUSTRY EXPERIENCE

R&D Engineer Intern — Beijing Zhangshangxianji Network Technology Co., Ltd.

Jun 2021 - Sep 2021

- Implemented Spring Boot transaction services for suppliers, retailers, and agencies.
- Improved system performance through unified Single Sign-On; built SMS authentication, OCR-resistant verification images, complex database access, and department-wide alerting with messaging and automated calls.

AWARDS, PRESENTATIONS & TEACHING

- Best Poster Award (Basic Science), NUHS Scientific and Innovation Summit, 2026.
- NUS Research Scholarship, 2022-2026.
- Invited talk at AP-CCN (Asia-Pacific Computational and Cognitive Neuroscience Conference).
- Posters at OHBM (Organization for Human Brain Mapping) 2026, NUHS Scientific and Innovation Summit 2026, and AI4X 2025.
- Teaching Assistant for EE2026 Digital Design and EE3731C Signal Processing at NUS.

TECHNICAL SKILLS

- **Programming:** Python, MATLAB, TypeScript/JavaScript, C, C++, Java
- **Machine Learning & Scientific Computing:** PyTorch, JAX, scikit-learn, SciPy; deep learning, surrogate modeling, numerical optimization, simulation
- **Statistical Methods:** Cross-validation inference, hypothesis testing, simulation-based validation, false-positive-rate/power/confidence-interval calibration, systematic review and meta-analysis
- **Neuroimaging Analysis:** Resting-state fMRI analysis, functional-connectivity analysis, brain parcellation, cortical-surface visualization; FreeSurfer/Freeview
- **AI Agent Systems:** Codex, Claude Code; agent workflow design, tool orchestration, structured outputs, provenance tracking, deterministic validation, human-in-the-loop review
- **Software & Web:** Node.js, React, Phaser, Vite, Zustand, Capacitor
- **Research Infrastructure:** Git, GitHub Actions, Linux/bash, HPC/PBS, conda, Docker
- **Languages:** Chinese (native); English (TOEFL 106, GRE 326 + 4.0)